Nodal-Precession Phasing for Multi-Object Debris Removal in Low Earth Orbit

An Audited and Reproducible Re-evaluation of Waiting below the Targets, with Exact Optimiser Checks and a Novelty Assessment

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Abstract

Massive derelict rocket bodies in a few low Earth orbit bands dominate the future fragment supply, and a single servicing spacecraft could visit many of them if it lets the J_2 nodal precession rotate the orbit planes into line instead of paying for plane changes. This paper re-derives, tests and re-runs an earlier draft study of that idea on one frozen public snapshot (Celestrak catalogue and element sets of 18 August 2026, common epoch 15:02:52 UTC, hashes published). It was found that several headline facts of the draft do not survive. The catalogue counts (63,553 rows, 1,288 actionable objects) cannot be reproduced, since the snapshot gives 1,151 actionable objects in the working band and 913 in 51 sub-clusters; the "lower is better" note rests on a Hohmann cost that is not symmetric; and the planner's scalarisation weight was never tuned. We prove that the drift-rate gain of lowering exceeds that of raising for every offset ($R(x) > 1$, series $1 + \frac{9}{2}x + \frac{81}{8}x^2 + \dots$), that the cost ratio is $1 + \frac{3}{2}x$ and that the net gain per ΔV is $1 + 3x$, and we give a decision rule, an eccentricity reversal and error bounds. Against an exact dynamic-programming optimum on 32 small instances, a tuned beam search is 1.9% (95% CI 0.6–3.4) from optimal, while the draft's configuration is 10.3% from it. On the 159-object SL-8 pool the median tour reaches 29 objects at 2 km/s and 7 yr, and 67 at 5 km/s and 15 yr, compared with 9 and 19 for impulsive plane changes and 6 and 12 for waiting without an altitude change. Cowell and SGP4 checks close the node to $\leq 0.05°$. Of ten novel ideas screened, only a parked carrier with expendable kits is supported (4.5–6 times faster kit placement than a serial tour); the rest are rejected, standard or prior art. The main drawback of the approach is that its advantage is bought with years of mission time, and we claim no new astrodynamics.

Nomenclature

Orbit and Perturbation Parameters	
a, e, i	semi-major axis [km], eccentricity, inclination [deg]
p	semi-latus rectum, $a(1-e^2)$ [km]
n	mean motion [rad/s]
μ, R_E, J_2	Earth gravitational parameter, equatorial radius, second zonal harmonic
Ω, $\dot{\Omega}$	right ascension of the ascending node (RAAN) [deg], secular nodal rate [deg/day]
K	nodal-rate constant, $\frac{3}{2}J_2R_E^2\sqrt{\mu}$
Phasing Parameters	
x	relative altitude offset, $\Delta a/a_T$
$R(x)$	ratio of nodal-rate gain for lowering to that for raising
g, g^*	nodal gap [deg], gap threshold of the decision rule [deg]
h	drift-orbit depth below (or above) the target [km]
Δt	dwell time to close a nodal gap [d]
Mission Parameters	
ΔV	velocity change [m/s]
v, v_c	orbital speed, circular speed [m/s]
I_{sp}	specific impulse [s]
C_dA/m	ballistic parameter [m^2/kg]
$w_{\Delta V}$, w_t	planner scalarisation weights on ΔV and time
K (kits)	number of expendable kits placed by the carrier

1 Introduction

Collision-risk studies since Kessler and Cour-Palais [17] and the long-term projections of the space agencies [22–25] agree that massive intact derelicts, above all rocket bodies concentrated in a few inclination–altitude bands, dominate the future fragment supply [1, 28]. Active removal is expensive per object, so one launch should remove many. The obstacle is the orbit plane. Objects in one band share inclination but not right ascension of the ascending node (RAAN), and a direct plane change of $\Delta\Omega = 1^\circ$ at $i = 83^\circ$ costs 127.5 m/s at the SL-8 orbital speed; Braun et al. [3] identify plane change as the main obstacle to multi-target removal. The remedy, already used in constellation deployment [9, 18, 27] and in debris tours [5, 12, 31], is to let the secular J_2 drift of the node do the work: the spacecraft waits on an orbit whose nodal rate differs from that of the target.

This paper started from a draft study that proposed waiting *below* the targets (lowering) and reported tours of 24–53 objects. However, the draft's numbers were never independently reproduced, its analytic claims were not checked, and its planner was never compared with an exact optimum. The goal of this paper is to address this gap. This study turns the draft into a reproducible evaluation to a publishable standard, and tests by calculation whether further ideas help. The contributions are evidential, not conceptual (Section 10.2):

1. an independent engine that reproduces the draft's tour exactly, and an audit of every analytic claim and pipeline stage (Section 3);
2. a frozen, hash-pinned snapshot that resolves the catalogue-count discrepancies;
3. exact and sharp statements on lowering versus raising (Section 4);

4. an exact-optimum validation of the sequencing heuristics and a correction of the planner's weight (Section 6);
5. fair baselines, high-fidelity verification, Monte Carlo uncertainty, a caveated collision-risk benefit and a screen of ten novel ideas.

The rest of the paper is structured as follows. Section 2 develops the approach: the secular nodal-rate and cost models, the frozen snapshot, the leg and tour model, the solvers, the baselines and the uncertainty and risk models. Section 3 reports the audit of the draft, Section 4 the exact results on lowering versus raising, and Sections 5 and 6 the validation of the numerical model and of the optimiser. The results are then presented for the fair baselines, the headline missions and the collision-risk benefit (Section 7), followed by uncertainty and sensitivity (Section 8) and the screen of novel methods (Section 9). The discussion covers the implications, the novelty assessment, the sources of error and model limitations, and the objections a reviewer would raise (Section 10). The paper concludes with the main findings and the work that remains.

2 Approach

The goal of this study is to quantify how many objects a single chaser can visit in one band when the plane changes are done by J_2 nodal drift, and how that compares with the alternatives at equal budget. The analysis links

1. a secular nodal-rate model and exact transfer costs (Section 2.1),
2. a frozen catalogue snapshot and a leg model that turns each target visit into a transfer, a dwell and a final transfer, and
3. a tour optimiser checked against an exact dynamic program,

to compute the number of objects reached, the ΔV, the mission time and the propellant mass, together with their uncertainty.

2.1 Secular nodal rate and transfer costs

Secular nodal rate. For semi-major axis a, eccentricity e, inclination i and $p = a(1-e^2)$, the J_2 secular rate is [4, 19, 20]

$$\dot{\Omega} = -\tfrac{3}{2} J_2 n \Big(\frac{R_E}{p}\Big)^2 \cos i, \qquad n = \sqrt{\mu/a^3}, \tag{1}$$

where n is the mean motion, μ the Earth's gravitational parameter and R_E its equatorial radius. For $e = 0$ the rate reduces to $\dot{\Omega} = -K \cos i \; a^{-7/2}$, with $K = \frac{3}{2} J_2 R_E^2 \sqrt{\mu}$. A target whose rate is higher than the chaser's, in the direction of the gap, is reached after $\Delta t = g/|\dot{\Omega}_c - \dot{\Omega}_t|$ for a gap g. For absolute rates we use the second-order closed form (terms in J_2^2 and J_4) with the SGP4 constants, which reproduces SGP4's own mean nodal rate to 0.08%; Eq. (1) alone is 0.29% high on the SL-8 pool. However, only rate *differences* matter for phasing, so this common-mode bias cancels.

Costs. Hohmann transfers use the exact two-burn expression. A node error of θ_Ω at inclination i costs $2v \sin(\frac{1}{2}\vartheta)$, with $\sin\frac{1}{2}\vartheta = \sin i \, \sin\frac{1}{2}\theta_\Omega$ (equal inclinations). Drag make-up is computed with a tabulated exponential atmosphere and a ballistic parameter $C_d A/m = 0.01\,\mathrm{m}^2/\mathrm{kg}$ (stated, unverified). Propellant follows the rocket equation for two stated designs, both of which are assumptions: a storable bipropellant design ($I_{sp} = 320\,\mathrm{s}$, 700 kg dry) and a Hall-thruster design (1800 s, 900 kg dry).

2.2 Frozen snapshot and pools

The analysis uses *snapshot v3-2026-08-18*, which consists of the Celestrak SATCAT csv (70,292 rows) and the GP element sets for five pools, fetched by the original study on 18 August 2026 from public, credential-free sources; the SHA-256 hashes are in `data/SNAPSHOT.json`. The snapshot was not re-downloaded, because the draft's tours are defined on it and a new epoch would break the comparison. All elements are propagated with SGP4 to one common epoch (2026-08-18T15:02:52.746Z for the SL-8 set; TLE age median 0.14 d, maximum 0.87 d). The pools are SL-8 rocket bodies ($i \approx 82.9°$, 950–1,010 km, 159 objects), sun-synchronous objects (135), Iridium-class objects (28) and SL-16 rocket bodies (20). Nodal rates are evaluated on *mean* elements. The draft used osculating elements, which differ in a by 6.4 km (1σ), a difference comparable to the pool's own spread of 7.5 km.

2.3 Leg model and tour problem

A leg takes the chaser from object c (in its plane) by a Hohmann transfer to a circular drift orbit $a_d = a_t \mp h$ below (or above) target t. The chaser then dwells until the secular nodal gap closes, and finally makes a combined Hohmann and inclination transfer to t, plus a fixed proximity-operations allowance (15 m/s, stated). The depths considered are $h \in \{25, 35, 50, 75, 100, 150, 200, 300, 400, 500, 600, 700\}$ km (altitude ≥ 200 km), with dwell ≤ 2200 d and relative rate $\geq 0.01°$/day. After a leg the chaser lies in the target's plane, so its node depends only on (object, epoch). The tour is therefore a *time-dependent orienteering problem*: maximise the number of objects visited subject to a ΔV budget and a mission-time cap, where the arc cost depends on the departure epoch through the evolving gaps. Two envelopes are taken from the draft: "small" (2 km/s, 7 yr) and "large" (5 km/s, 15 yr). The counts include the free start object, since launch is assumed to insert the chaser at the start object. We report distributions over *all* start choices rather than the best of N.

2.4 Solvers and the scalarisation weight

Greedy and beam search rank actions by $w_t\,\Delta t/100\,\mathrm{d} + w_{\Delta V}\,\Delta V/100\,\mathrm{m/s}$. The draft used $w_{\Delta V}{:}w_t = 10{:}1$ and never tuned it. With both budgets binding, we find the Lagrangian-consistent weight by a sweep on random starts and report it on disjoint held-out starts (Section 6). We also test a rollout beam and a permutation NSGA-II (three seeds). The exact reference is a Held–Karp dynamic program over (visited set, current object) with Pareto labels (time, ΔV). It is exact for this leg model up to the weak departure-epoch dependence of drag make-up, and it is checked by brute force on tiny instances (`tests/`).

2.5 Baselines

All baselines use the same pool, leg costs and budgets. The metrics are objects reached, ΔV, time, and propellant mass by the rocket equation.

- *Impulsive*: direct plane-change hops, $2v\sin(\vartheta/2)$ plus Hohmann plus 15 m/s, hop time 2 d, beam width 6.
- *Natural drift only*: no altitude change; the chaser keeps its own orbit while waiting.
- *Published-style drift phasing*: one fixed raised waiting orbit at +40 km (the single raised orbit of [31]), and one fixed waiting orbit at −100 km (the carrier below the targets of [8]).
- *Free per-leg depth, lowering only* (this study).
- *Free per-leg depth, raising or lowering*, with one extra synodic lap allowed (the intermediate drift orbit "below, between or above" of [5]).

2.6 Uncertainty and verification

Sampled legs of the best tour are propagated numerically (Cowell, J_2–J_4, drag, solar radiation pressure, Sun and Moon) with SGP4 targets, and the node closure error is measured. A Monte Carlo then draws closure errors, calibrated on that check, with element-set growth, burn execution, density and lognormal proximity-operations variability, and counts the objects completed within the budget. A one-at-a-time sensitivity study uses paired differences over fixed starts.

2.7 Collision-risk benefit

A kinetic-gas model with latitude-resolved spatial density gives the catastrophic-collision involvement rate of each object against the catalogue population. The cross-section is the sum of radii, a collision is catastrophic if $\frac{1}{2}m_s v^2/m_l \geq 40\,\mathrm{J/g}$, and mass is obtained from the radar cross-section (RCS) by an imputed relation, with 66% of radar cross-sections imputed. Fragment yield follows the standard breakup power law [15]. The model is a ranking and scaling tool with large uncertainty and is *not* a prediction; the draft's superseded "95.4%" claim is not re-derived.

2.8 Key assumptions

The main modelling assumptions are summarised below; their effect is discussed in Section 10.3.

- Legs are impulsive and circular-to-circular, with no in-plane phasing; proximity operations and capture are one fixed allowance of 15 m/s.
- Launch inserts the chaser at the start object at no cost.
- Drag make-up uses a tabulated exponential atmosphere and $C_d A/m = 0.01\,\mathrm{m^2/kg}$.
- The dry masses and specific impulses of the two propulsion designs, and the kit mass, are stated assumptions rather than sourced values.
- The node uncertainty at epoch (0.01°) is an assumption, since no published RAAN uncertainty was found.

3 Audit of the Draft Study

Table 1 lists the findings that change a headline or a stated claim; the full list (13 analytic and 13 pipeline items) is in `audit/AUDIT.md`. The independent engine reproduces the draft's own tour (29 objects, 1,445.48 m/s, 2,555.54 d, node closure 2×10^{-13} deg at all 28 legs), so the differences between the two studies are model or input choices.

4 Lowering versus Raising: Exact Results

We first establish the exact comparison between waiting below and waiting above a target. Let $x = \Delta a/a_T \in (0,1)$ and $f(a) = a^{-7/2}$.

Theorem 1 (Rate gain)**.** *For circular orbits with the same inclination and J_2 only, $R(x) \equiv \dfrac{|\Delta\dot{\Omega}_{\downarrow}|}{|\Delta\dot{\Omega}_{\uparrow}|} = \dfrac{(1-x)^{-7/2}-1}{1-(1+x)^{-7/2}} > 1$ for every $x \in (0,1)$, independent of i.*

Proof. f is strictly convex, so $f(a-\Delta)+f(a+\Delta) > 2f(a)$, i.e. $f(a-\Delta)-f(a) > f(a)-f(a+\Delta) > 0$. The factor $K\cos i$ cancels. □

Table 1: Audit Findings That Change a Headline (Severity A) or a Stated Claim (B)

	Draft claim	Finding
A	63,553 rows → 1,288 actionable objects; 40 sub-clusters, 593 objects	Not reproducible on any cut of the snapshot. Strict rule, 600–1,500 km: 1,151; clustering gives 51 sub-clusters, 913 objects. 63 of the 593 shipped rows carry decay dates before 2025; only 270 carry a RAAN
A	Hohmann cost symmetric; net advantage $1+4.5x$	$\Delta V_\downarrow/\Delta V_\uparrow = 1 + \frac{3}{2}x + \frac{9}{8}x^2$; net per-$\Delta V$ gain $1 + 3x$
A	Efficiency has an interior optimum 150–300 km below	No interior optimum (gain per ΔV rises monotonically); quoted values about 11 times too large
A	"59.8% saving" over plane changes	No time-free saving exists. At 1° the saving is 80% if 279 d are allowed, 13% if 30 d. Same-sequence replacement on the draft tour: 72.5%
A	Headline counts (24, 29, 53 objects)	Best-of-60 starts; over all 159 starts the draft's configuration has median 20 (CI 19–20), maximum 28
A	Weights 10:1	Far from the Lagrangian optimum; counts understate the same model by 45–90%
A	"Objects removed"	The tours *visit*; removal needs one kit per object (Section 9)
B	Alignment window 1°	Defined by cost: 12.8 m/s per 0.1°; the 15 m/s allowance covers 0.118°
B	$R > 1$ "for every target"	False for eccentric targets (reversal near $e \approx 0.03$)
B	Greedy and differential evolution	Results come from greedy, beam and NSGA-II; DE exists in the repository but is not behind the numbers
B	Elements	Osculating inputs: nodal-rate noise 0.0026°/day (1σ) against a pool spread of 0.0066°/day

Proposition 1 (Series and cost)**.** $R = 1 + \frac{9}{2}x + \frac{81}{8}x^2 + \frac{141}{8}x^3 + O(x^4)$*, with* $0 \le R - 1 - \frac{9}{2}x \le 13.58\,x^2$ *for* $x \le 0.15$ *(supremum checked on a 20,000-point grid and symbolically). The exact Hohmann costs satisfy* $\Delta V_\downarrow = \frac{1}{2}x + \frac{3}{8}x^2 + \dots$ *and* $\Delta V_\uparrow = \frac{1}{2}x - \frac{3}{8}x^2 + \dots$ *(units of* v_c*), hence* $\Delta V_\downarrow/\Delta V_\uparrow = 1 + \frac{3}{2}x + \frac{9}{8}x^2 + O(x^3)$*, and the gain per unit* ΔV *is larger for lowering by* $R\,\Delta V_\uparrow/\Delta V_\downarrow = 1 + 3x + O(x^2)$*.*

At $a_T = 7350$ km (975 km altitude), $R = 1.031, 1.063, 1.130$ and the equal-ΔV gain ratio is $1.021, 1.042, 1.087$ for offsets of 50, 100 and 200 km (first-order $1 + 3x$: 1.020, 1.041, 1.082). The error order $O(x^3)$ stated in the draft is therefore wrong, since the remainder is $O(x^2)$. Figure 1 shows these results.

Both signs, inclination and eccentricity. For retrograde orbits ($\cos i < 0$) the rates change sign and "lowering" accelerates the drift eastward; Theorem 1 is unchanged. Including J_2^2 and J_4 changes R by at most 1.1×10^{-3} across $i = 52$–$98.7°$. A circular chaser against an *eccentric target* sees an effective altitude offset, because $\dot\Omega_t \propto a^{-7/2}(1-e^2)^{-2}$ is the rate of a circular orbit lowered by $\frac{4}{7}e^2 a$ (0.1 km at $e = 0.005$, 10.5 km at 0.05, 42 km at 0.1). For a 100 km offset, $R = 1.063, 1.054, 1.028, 0.861, 0.433$ at $e = 0, 0.01, 0.02, 0.05, 0.1$, so raising wins above $e \approx 0.03$. This is irrelevant to the SL-8 pool ($e \approx 0.002$) but affects 62 of the 593 shipped objects.

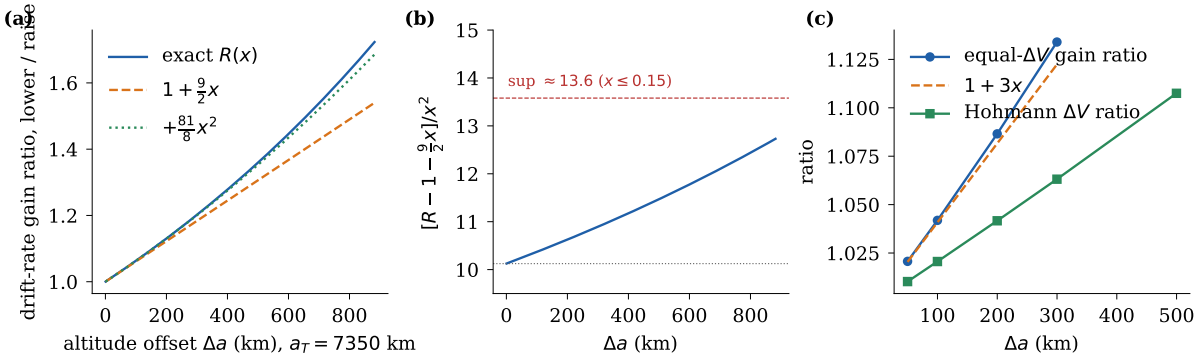


Figure 1: Lowering versus raising for circular orbits. (a) Exact rate-gain ratio $R(x)$ and its series. (b) Scaled remainder, bounded by 13.58. (c) Hohmann cost ratio and equal-ΔV gain ratio against $1 + 3x$. Lowering wins in both rate and cost, but by only a few per cent at practical offsets.

Drag. The $1+\frac{3}{2}x$ cost ratio applies to the transfer alone. When the dwell make-up is added, the lower-to-raise ΔV ratio at 975 km is 1.009 (50 km offset, mean activity); near 600 km it is 1.1 (mean) to 1.4–1.5 (high activity). The draft's "2 km/day drag floor" never binds (decay at 300 km altitude is 0.36 km/day); the effective limits are the 200 km altitude floor and the make-up cost (about 1 m/s per day at 275 km).

Decision rule. Lowering only closes gaps of one sign (westward for prograde orbits). For a gap $g \in (0, 360°)$ in the lowering direction, lowering is faster when $g < g^* = 360R/(1 + R) \approx 180°(1 + \frac{9}{4}x)$; the approximation errs by $\le 0.46°$ for $x \le 0.1$. Allowing both senses halves the expected wait for a uniform random gap (Monte Carlo ratio 1.94, theory 2), and on SL-8 pairs the rule picks the faster sense in 94–96% of cases (wait ratio 2.15 at 50 km). The "always lower" design rule is therefore a weakness rather than a result, and the bidirectional planner is the right baseline.

5 Validation of the Numerical Model

Secular rate. Against a J_2-only Cowell integration, Eq. (1) evaluated on mean elements is within 0.04–0.08%; with the osculating a it is 0.34–0.46% off. Adding J_3 and J_4 shifts the rate by 0.02–0.23%, an effect dominated by J_4 (Fig. 2).

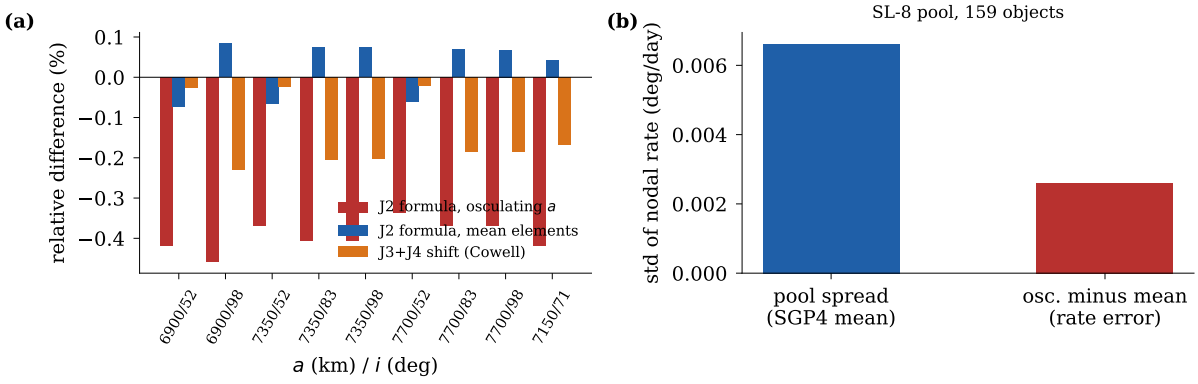


Figure 2: Validation of the secular nodal rate. (a) Secular-formula error against Cowell integration for nine (a, i) cases. (b) Pool spread of the nodal rate against the rate error from using osculating elements; the osculating error is a significant fraction of the spread, which is why mean elements are used.

Leg closure. For eight legs of the tuned best tour, a circular Cowell chaser (full forces) and SGP4 targets give node closure errors of $-0.002 \pm 0.017°$ (mean $\pm$ s.d., maximum $0.039°$) for the first-order plan and $-0.006 \pm 0.018°$ (maximum $0.049°$) for the second-order plan. These correspond to about 1.3–1.5 m/s of correction per leg, against the 15 m/s allowed ($0.118°$). A J_2-only numerical model disagrees by $0.08°$ because the plan includes J_4; adding J_3 and J_4 closes this to $0.003°$, and drag, SRP and Sun/Moon account for the rest (Fig. 3). Planned and propagated dwell differ by up to 3 d. Eight legs is a small sample, and Section 10.3 states what is not covered.

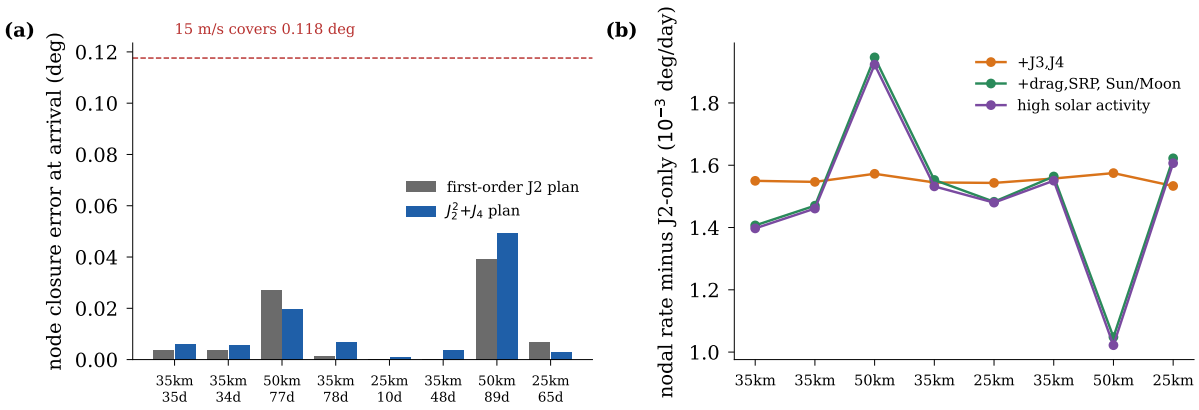


Figure 3: High-fidelity verification of the leg model. (a) Node closure error at arrival on eight sampled legs; the dashed line is the error covered by the 15 m/s allowance. (b) Nodal-rate increment of each added force model over J_2 only.

Code. Fifteen automated tests pass. They cover the zonal accelerations against the potential gradient; the series and bound of Proposition 1; the Hohmann ratio and the plane-change small-angle limit; the second-order rate against SGP4; the drag identity; the replay of the draft's tour; the closure and budget invariants; the dynamic program against brute force; the funnel counts; and the risk-model scaling.

6 Optimiser Validation

Weight. We swept $w_{\Delta V}$:w_t on random starts and evaluated the result on disjoint held-out starts. The best weights are 2 (small envelope) and 1.5 (large), against 10 in the draft, and the held-out medians are 31 versus 22 objects (small) and 68.5 versus 39.5 (large). Figure 4 shows the sweep. This is standard multiplier tuning, not a new technique; the point is that the draft's published counts understate its own model.

Distance from optimum. On 32 small instances (16 windows of 10 and of 12 consecutive-RAAN objects, with budgets of 700 m/s and 600 d, and of 800 m/s and 700 d) the exact dynamic program gives the optimum count. The mean gaps (95% bootstrap CI) are: greedy at the draft weight 18.7% (16.0–21.4); beam-20 at the draft weight 10.3% (8.0–12.7); NSGA-II 9.7% (7.1–12.2); tuned greedy 6.1% (4.1–8.2); tuned beam-20 1.9% (0.6–3.4), optimal in 81% of instances (worst 12.5%); and rollout beam 2.6% (1.0–4.2). Beam width saturates at about 10 on the full pool (best start 34, 34, 35, 35, 35, 35, 35 objects for widths 1–1000; Fig. 5). The ΔV-only relaxation bound is on average 31% above the exact optimum on the small instances. On the full pool it is 71 objects at 2 km/s (best found 36) and vacuous at 5 km/s (160 against a pool of 159), so the

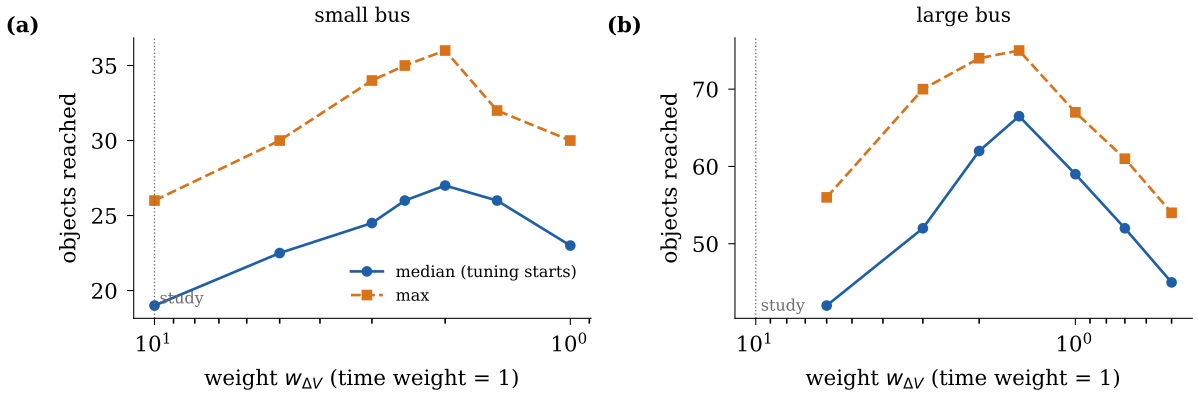


Figure 4: Objects reached against the scalarisation weight. The draft used 10, well away from the best values for either envelope.

full-pool gap is not certified and the small-instance gaps are the evidence. The draft's belief that wider beams or evolutionary search would help is not supported; the weight mattered more.

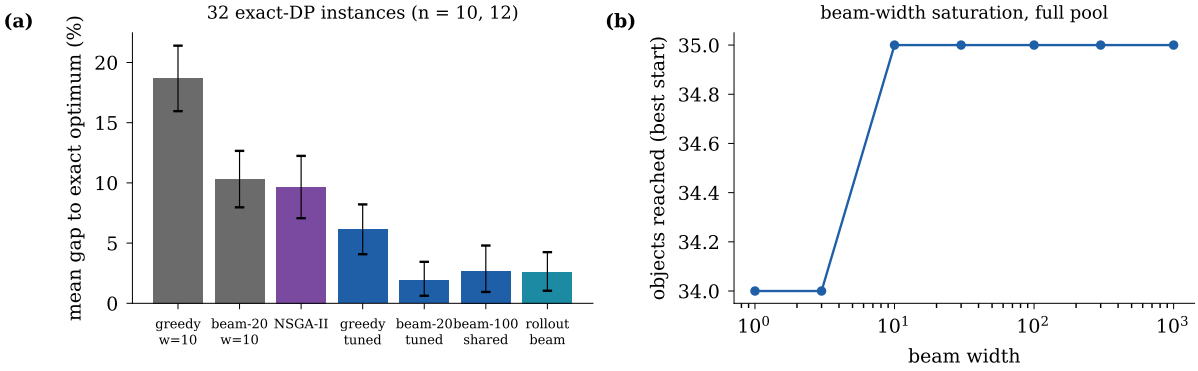


Figure 5: Optimiser performance. (a) Mean gap to the exact optimum with 95% CIs. (b) Beam-width saturation on the full pool. The tuned beam search is close to optimal, and widening the beam beyond about 10 gives no gain.

7 Results

7.1 Fair comparison with natural drift and impulsive plane changes

To place the drift tours in context, we compare them with every baseline at equal budget. Table 2 gives the medians over all 159 starts (bootstrap CI), with ΔV, time and propellant, and Figure 6 shows the full distributions. Three comparisons from Table 2 warrant discussion. First, at equal ΔV budget the drift tour reaches 3.2 times (small) and 3.5 times (large) as many objects as impulsive hopping, but it needs 2,500–5,300 d against 16–36 d. The saving is a trade of time for propellant, and the value of that trade depends on the bus life, not on a universal percentage. Second, waiting with no altitude change is cheap (114 m/s) but slow, reaching only 6 objects. Third, the published-style *fixed* waiting orbits do worse than free per-leg depth: the +40 km orbit reaches 16 (median) and the fixed −100 km orbit 18, against 29. The −100 km result is deterministic here, since every leg costs nearly the same and the budget binds. The gain over the published rules therefore comes from the per-leg choice of depth and sign. The bidirectional planner adds about two objects at 2 km/s and nothing measurable at 5 km/s.

Table 2: Objects Reached (Including Start), SL-8 Pool, All 159 Starts

Method	median	95% CI	max	ΔV (m/s)	time (d)	chem (kg)	EP (kg)
small envelope: 2 km/s, 7 yr							
draft as audited	20	19–20	28	1664	2555	489	89
v3 inputs, draft weight	20	19–22	27	1707	2554	506	91
fixed raise +40 km	16	14–17	30	851	2451	218	44
fixed lower −100 km	18	18–18	18	1977	1010	614	107
free lower, tuned	29	27–31	36	1968	2513	611	106
free bidirectional + laps	31	28–32	36	1971	2479	612	106
natural drift only	6	6–7	12	114	2267	26	6
impulsive plane change	9	8–10	15	1775	16	532	95
large envelope: 5 km/s, 15 yr							
draft as audited	35	32–38	52	2500	5477	853	137
fixed raise +40 km	32	29–36	51	1779	5329	534	95
fixed lower −100 km	44	44–44	44	4998	2732	2742	295
free lower, tuned	67	64–68	76	4976	5325	2718	293
free bidirectional + laps	67	67–70	77	4974	5342	2716	293
natural drift only	12	12–13	22	440	5138	105	23
impulsive plane change	19	18–22	31	4779	36	2510	280

Propellant (fixed-orbit rows: rocket equation on the median ΔV): storable chemical design (700 kg dry) and electric design (900 kg dry, 1800 s), rocket equation.

7.2 Headline missions on the frozen snapshot

Table 3 gives the headline re-run of the two envelopes with the tuned weights. The draft's 29 is

Table 3: Headline Re-run on the Frozen Snapshot

	2 km/s, 7 yr	5 km/s, 15 yr
tuned weight	2	1.5
median over all starts (95% CI)	29 (27–30)	67 (64–68)
greedy median	28	65
best start	36 (1995 m/s, 2397 d)	76 (4981 m/s, 5090 d)
best-of-60 starts, mean (CI)	35.3 (34–36)	75.9 (75–76)
Monte Carlo, best tour: refreshed elements	33.0	71.8
unrefreshed (open loop)	28.7	58.9
pessimistic errors	30.3	67.4
draft as published (best of 60)	29 (1445.5 m/s, 2555.5 d)	53 (4101 m/s)
four bands, medians summed (SL-8, SSO, Iridium-class, SL-16)	87 (best 101)	171 (best 188)

SL-8 pool, beam width 12 on all 159 starts unless stated; Monte Carlo of execution errors on the best tour.

reproduced exactly by the independent engine with the draft's settings, and it is the maximum over starts of a configuration whose median is 20. With the weight corrected, the median is 29 and the best 36. In the large envelope the pool is nearly exhausted by the best tours (76 of 159), so the counts there are pool-limited and are not a measure of the method's reach. Figure 7 shows the two best tours.

7.3 Collision-risk benefit

To connect the tours to the debris environment, we compute the catastrophic-collision involvement rate removed by each median tour (Table 4, Fig. 8). For the SL-8 median tour the removed hazard is 18% of the pool's total (0.0088 events/yr). Over 25 years the model gives 0.040 catastrophic collisions avoided (Monte Carlo 5–95%: 0.025–0.055) and about 65 fragments > 10 cm (21–217),

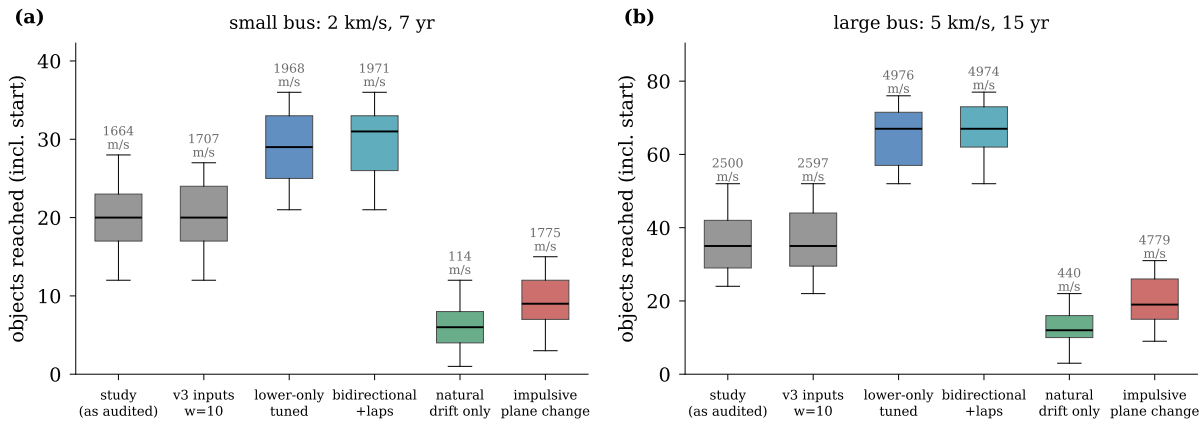


Figure 6: Objects reached over all 159 starts for each method (box: quartiles; whiskers: range); labels give the median ΔV. Free per-leg depth reaches the most objects, while impulsive hopping is the only fast option.

Table 4: Catastrophic-Collision Involvement Rate Removed by the Median Tour (Kinetic-Gas Model; *Ranking-Level Accuracy Only*)

band	pool	tour n	tour rate (10^{-3}/yr)	top-n	random	CV
SL-8	159	29	1.58	1.85	1.61	0.16
SSO	135	27	1.27	2.69	1.31	0.67
Iridium-class	28	11	0.55	0.65	0.53	0.27
SL-16	20	16	0.87	0.89	0.87	0.10

Tour n is the median object count including the free start; the benefit calculation below uses the median-hazard tour (28 objects for SL-8). CV: coefficient of variation of per-object rate; "top-n": removing the n highest-rate objects; "random": random n.

while a random set of the same size gives 0.039. In SL-8 the per-object hazard is nearly uniform (CV 0.16), so *risk-aware selection is worth nothing* there. In the sun-synchronous band it would double the benefit (CV 0.67), but the tour as sequenced captures only half of what the top-n set would. Choosing the start by risk raises SL-8 from 28 to 36 objects and the benefit from 0.040 to 0.050. The cost per avoided event is a scaling statement, not an estimate: at an assumed \$100–400 M per mission, the nominal figure is \$2.5–10 B per avoided catastrophic collision (\$1.8–16 B across the 5–95% interval of avoided events). Published cost anchors are sparse and are not used as inputs. The result must not be read as a case for or against removal, because the model has no fragment-cascade feedback beyond the first generation (amplification 1.06) and the long-term benefit of avoiding the *first* fragmentation in a band is not captured.

8 Uncertainty and Sensitivity

Execution errors. The closure-error Monte Carlo (calibrated on Section 5) yields, for the planned 35-object tour at the 2 km/s budget, 32.0 objects completed with an element refresh at each leg (a receding-horizon replan), 27.8 without refresh (the 95th percentile node error is then 0.36°, three times the 15 m/s allowance) and 28.8 under pessimistic errors. A planning margin of 0–15% of the budget does not help the open-loop case (27.5–27.9 at every margin), so the recovery comes from refreshing the elements and not from reserve (Fig. 9). The first-order draft model and the second-order model give nearly the same count (32.1 and 32.0), because the rate bias is common mode.

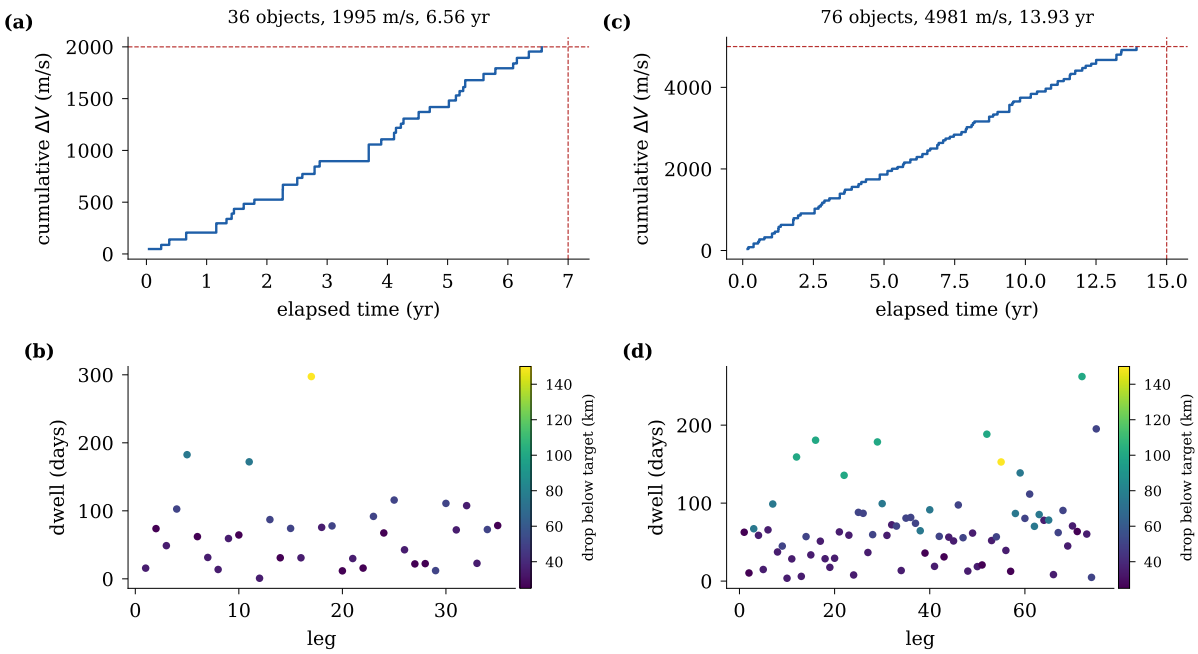


Figure 7: Best tours for the two envelopes. (a,c) Cumulative ΔV against time; (b,d) dwell per leg, coloured by depth below the target.

One-at-a-time sensitivity. The sensitivity study uses paired differences over fixed starts (baseline 28.1 objects, Fig. 10). The largest losses come from a relative-rate floor of 0.03°/day (−9.8, CI −11.1 to −8.5); the draft weight of 10 (−8.9); a proximity-operations allowance of 50 m/s (−8.2) or 30 m/s (−3.7); an in-plane phasing allowance of 30 m/s and 20 d per leg (−7.3); a 5 yr life (−5.6); and a 1.5 km/s budget (−5.5). The gains come from a 3 km/s budget (+2.6), a 10 yr life (+1.8) and a 5 m/s allowance (+1.6). The ballistic parameter (0.003–0.03), solar activity, the decay cap, an altitude floor of 200 km, the leg-wait cap and inclination charging have exactly zero effect. The drag insensitivity is a property of this pool (drops of 25–150 km below 975 km, where decay is negligible) rather than a general result, and it would not hold for bands near 600 km. In-plane phasing is not otherwise modelled (closing 90–180° over 20 d costs 4.5–9 m/s round trip, and 18–36 m/s over 5 d); this is the most important unmodelled cost.

Beyond secular J_2. At 83° the absolute luni-solar nodal rate is 6×10^{-5}°/day, and its *differential* across a 100 km drop is 1.2×10^{-6}°/day, or 3×10^{-5} of the J_2 differential. With drag, SRP and third bodies, the full Cowell model differs from J_2–J_4 by 1×10^{-4}–4×10^{-4}°/day (below 0.05%). The SGP4 element-set error is the dominant data uncertainty: the TLE range error grows by about 1.5 km/day [21], no published RAAN uncertainty was found, and the node uncertainty at epoch (0.01°) is an assumption.

9 Novel Methods

Each idea is simulated against the baselines of Table 2 on the SL-8 pool with fixed seeds; the objective is the number of objects reached or placed, ΔV, time, and launch or propellant mass. The verdicts use only the numbers, and "prior art" cites what Section 10.2 found. Table 5 summarises the screen and Figure 11 shows the carrier comparison.

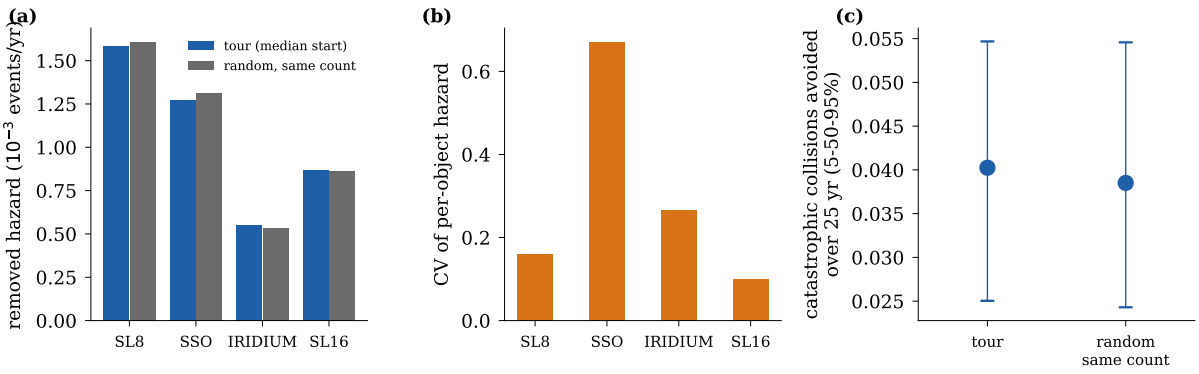


Figure 8: Collision-risk benefit of the median tours. (a) Removed hazard of the median tour and of a random set of equal size. (b) Heterogeneity of per-object hazard. (c) Collisions avoided over 25 yr. Ranking-level accuracy only.

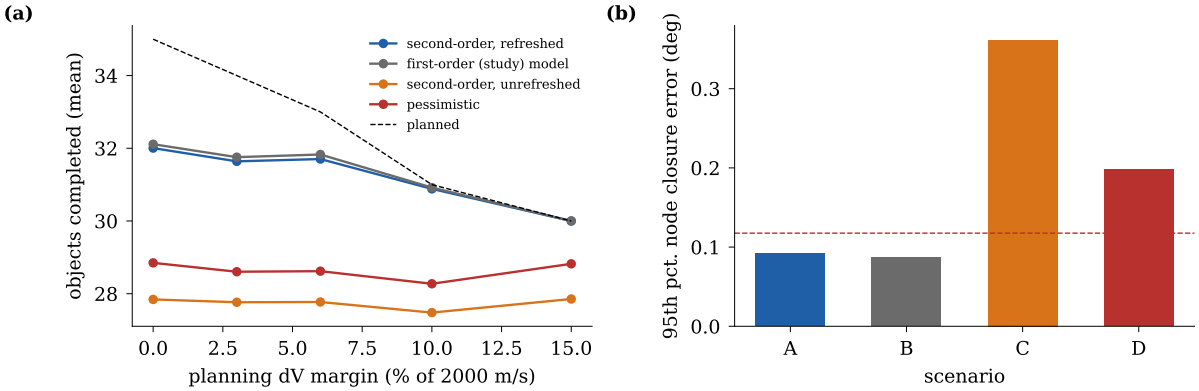


Figure 9: Monte Carlo of execution errors. (a) Objects completed against planning margin for four scenarios. (b) 95th percentile closure error (A: second-order refreshed; B: first-order draft; C: unrefreshed; D: pessimistic; dashed: error covered by the allowance). Refreshing the elements, not adding margin, recovers the lost objects.

The one supported idea. A carrier parks on a *single persistent* lowered drift orbit (depth 400 km, $\Delta V \approx 210$ m/s including make-up) whose plane sweeps through the cluster. When the node of its orbit coincides with that of a target within the tolerance, a kit departs (kit ΔV 260 m/s, 5 days, 30 m/s extra), and the carrier does not return to any target plane. The K nearest-in-time targets are served in the order in which the sweep reaches them. Depth trades time against launch mass (Fig. 11), and the time CI is a bootstrap over all 159 starts. The comparison fixes the kit mass at 135 kg (unverified), counts kit propellant with $I_{sp} = 280$ s, and omits kit avionics, capture hardware and rendezvous autonomy. The launch-mass advantage (5% at $K = 8$, 20% at $K = 16$, 50% at $K = 36$) is therefore an upper bound. The time advantage (4.5 to 6.3 times) is less sensitive: kit allowances of 60–100 m/s and 10–20 d change the median at $K = 16$ from 298 to 313 d at 300 km depth. The hopper medians use 20 sampled starts. Prior art covers a carrier with kits that waits in a drift orbit [7, 8, 31]; whether kits depart from a persistent orbit without a carrier return was not found in the sources reached, but absence in a bounded search is not proof.

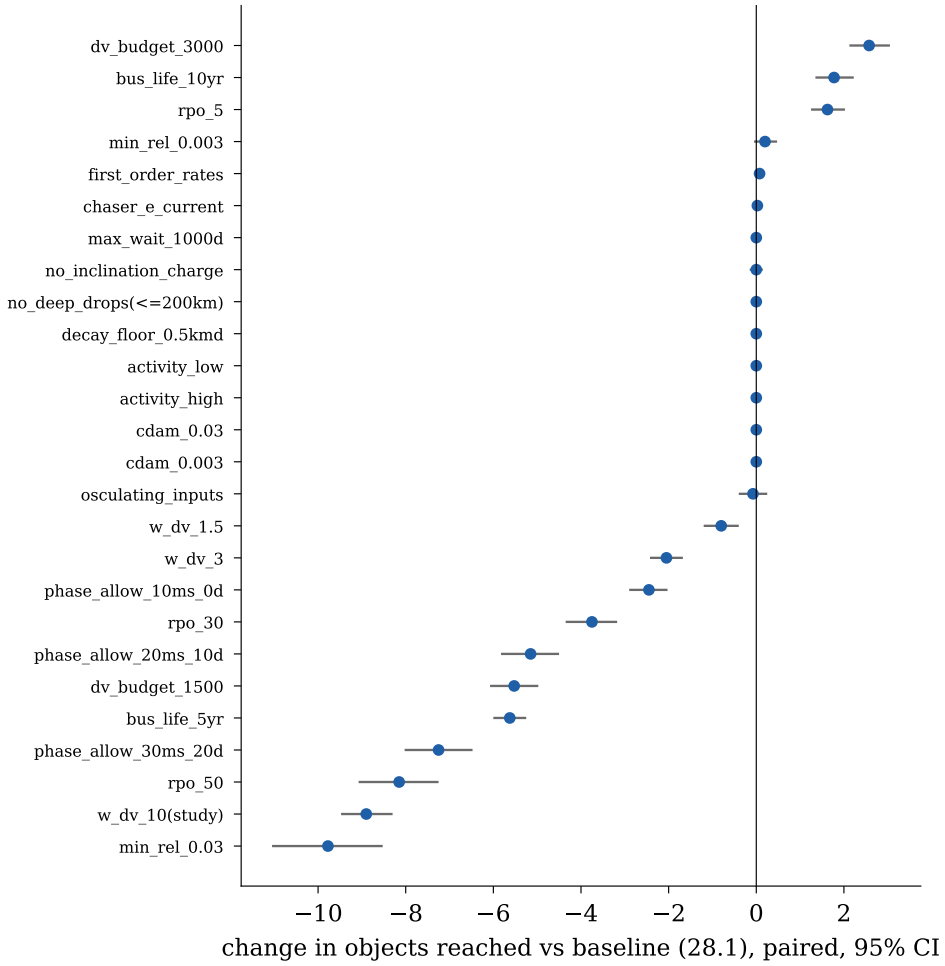


Figure 10: One-at-a-time sensitivity: change in objects reached against the baseline, paired, with 95% CI. The proximity and phasing allowances dominate; drag has no effect for this pool.

10 Discussion

10.1 Implications for multi-object removal

The physical core is simple. The nodal rate is a steep function of altitude, so a drop of 25–150 km buys rate differences of 0.01–0.06°/day for 25–150 m/s round trip, and the node of a prograde orbit at 83° then closes a few-degree gap in 30–300 d. The numbers add five points to the draft. First, the size of the advantage over impulsive plane changes is a function of the time allowed, and it is large only if years of bus life are available. Second, the benefit of choosing the sign per leg is real (about two objects, and a factor of two in expected wait), so the unidirectional rule is not required. Third, the sequencing heuristic was not the weak point; the scalarisation weight and the best-of-N reporting were. Fourth, the visit count is not a removal count. With 135 kg kits, eight kits imply about 2.1 t at launch for a hopper, so removals per launch are mass-limited well below 29 or 67 objects, which is why the carrier-and-kits architecture is the one idea that matters. Fifth, in this model the collision-risk benefit of a 28-object removal from SL-8 is a few hundredths of a catastrophic collision over 25 years. The case for removal therefore rests on the heavy tail and on cascade dynamics, not on the first-generation average, and the model cannot settle it.

Table 5: Screen of Ten Ideas plus the High-Fidelity Check (EP: Electric Propulsion)

Idea	Result	Cost	Verdict
Continuous low-thrust profiles	Objects 25.5–28 (CI 24–31) against 28 impulsive; propellant 97–105 kg against 610 kg, entirely from I_{sp} (1800 s against 320 s)	thruster, power, time	no trajectory gain; prior art [5, 16]
Eccentric drift orbit / apsidal	Gain per ΔV 0.970–0.998 of circular, falling with depth	none	rejected
Oberth-type plane change at high apogee	Direct plane change cheaper than the bielliptic option for gaps up to 30° (127.5 against 163 m/s at 1°); drift 25–55 m/s for 1–10° in 2 yr	–	rejected for small angles
Luni-solar plus J_2	Differential rate 3×10^{-5} of J_2	–	rejected
Drag sail as altitude shaper	Time to descend 100 km for C_dA/m = 1–0.1 m^2/kg, mean activity: 16–156 d at 500 km, 77–767 d at 600 km, 330–3,270 d at 700 km, 840–8,370 d at 775 km, 4,700–47,000 d at 975 km (high activity 5–9 times faster)	sail area and mass; no control of re-raise	useful below about 700 km; impractical at 975 km
Deorbit of removed object to retime	No mechanism: the removed object leaves the pool, the chaser's orbit is unchanged. Towing a 1,435 kg object 100 km down costs 50.6 m/s, triples the propelled mass, and does not deorbit it at 975 km	–	rejected
Better tour formulations	Tuned weight: +9 objects (small), +32 (large) median; rollout beam and exact-gap results (Section 6); receding-horizon refresh +4.2 objects (Section 8)	compute only	supported, but standard
Two or three servicers, one cluster	Same total ΔV split 1/2/3 ways: 35/36/39 objects, time 2493/1380/887 d, propellant 621/513/477 kg; k full servicers: 35/65/94	more hardware	time and mass benefit, no count benefit
Band choice by RAAN concentration	40-object pools: RAAN-window 21, random 16, evenly spread 14 objects	–	supported (pool design)
Parked carrier + expendable kits	K = 16: placement 219 d (CI 195–246) against 1198 d for a hopper; launch mass 3.3 t against 4.1 t; kit $\Delta V \leq 261$ m/s; K = 36: 484 d against 3041 d, 6.5 t against 13.4 t	kit autonomy and kit bus mass (not counted)	**candidate**; partial prior art
(verification) Full Cowell model	Full Cowell against J_2–J_4: slope differences $\leq 4 \times 10^{-4}$°/day	–	supports the model

10.2 Novelty assessment

Already known. Using J_2 drift for multi-debris tours, with drift orbits below, between or above the debris, is in Cerf (2013), with a lower-orbit worked example [5, 6]. The GTOC9 winners exploited it in branch-and-bound and beam searches [13, 30]. Raised waiting orbits with deployable kits are in [31], and a carrier with 2–8 kits waiting in a drift orbit 100 km below is in [8]. That altitude-lowering is more efficient than raising for nodal phasing is in [27], and the principle is patented prior art from 1993 [18]. Drift-orbit optimisation with drag in servicing is in [2], and with low thrust in [16]. Time-dependent TSP, evolutionary multi-chaser and dynamic-programming sequencing are in [10, 14, 26, 32]; risk-based ADR planning exists [11, 29]; and a recent preprint treats the drift orbit as a design variable and finds that inclination tilt beats altitude for near-sun-synchronous targets [12].

Not found in print (elementary, apparently unpublished). The closed-form ratio $R(x)$ with its remainder bound, the $1 + \frac{3}{2}x$ Hohmann ratio and $1 + 3x$ net gain, the decision threshold $180°(1 + \frac{9}{4}x)$ and the eccentric-target reversal. These follow from the convexity of $a^{-7/2}$ and small-parameter series, so their weight is small.

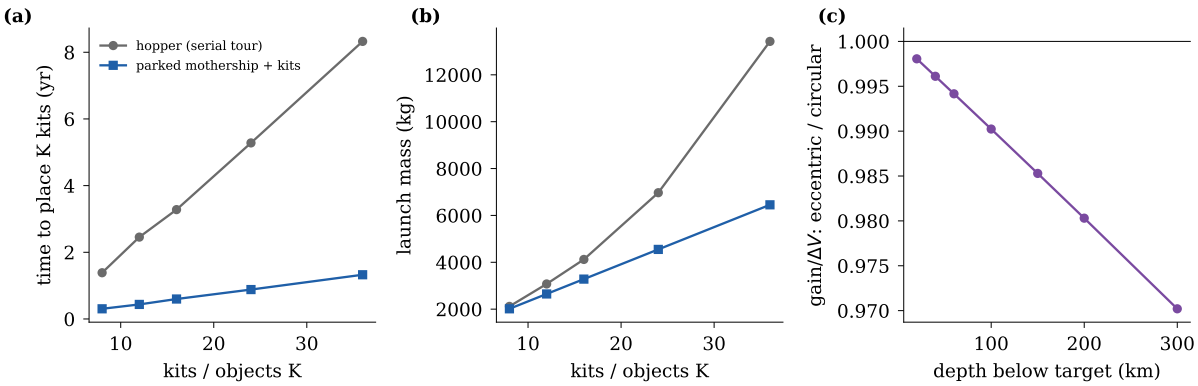


Figure 11: Novel-method comparisons. (a) Time to place K kits: serial hopper against a parked carrier at 400 km depth. (b) Launch mass. (c) Eccentric against circular drift orbit, gain per ΔV. The carrier places kits several times faster; the eccentric drift orbit gives no gain.

New evidence, not a new method. The audit and frozen snapshot; the weight correction and exact-optimum validation; the Cowell/SGP4 closure test; the quantified baselines; the risk benefit with its uncertainty; and the novel-idea screen (ten ideas).

Candidate novelty. The persistent-drift-orbit carrier whose kits depart without a carrier return, quantified here but untested against a full kit design. We do *not* claim new astrodynamics, any removal capability, or any prediction of environment evolution.

10.3 Sources of error and model limitations

Several simplifying assumptions in this analysis warrant discussion regarding their effect on the results.

Leg model: The leg model is impulsive and circular-to-circular and has no in-plane phasing. Proximity operations and capture are one fixed allowance (15 m/s), which is unverified and decisive: an allowance of 50 m/s costs −8 objects.

Free insertion: Launch inserts the chaser at the start object for free, whereas a real launch inserts at its own plane and node.

Element-set errors and verification sample: Element-set errors beyond the assumed values are untreated, and the verification covers eight legs of one tour.

Atmosphere: The atmosphere is a tabulated exponential model (unverified, uncertain by a factor of 2–3). Solar activity is bracketed, not simulated.

Mass and kit assumptions: The mass and kit assumptions (700 kg, 135 kg, I_{sp}) are stated, not sourced.

Risk model: The risk model is a kinetic-gas estimate with 66% imputed cross-sections, no cascade feedback and no operational-asset weighting.

Pools and epoch: The pools are four bands of a 2026-08-18 catalogue. Results for other bands (especially near 600 km, where drag matters) are not claimed. No re-fetch of the catalogue was made, so the numbers describe that snapshot only.

Relaxation bound: The relaxation bound on the full pool is loose, so the full-pool optimality gap is not certified.

Literature coverage: The literature coverage is a bounded search, and items marked unverified in `research/sources.md` were not used.

10.4 What a reviewer would challenge

The strongest objections to this analysis, and our answers, are as follows.

1. *"Your gain over the draft is a weight change."* Yes: the draft's configuration was untuned. The comparison against the *published-style* fixed orbits (Table 2) is the relevant one for a literature claim.
2. *"The impulsive baseline is a straw man."* It is the best single-burn hop with an exact Hohmann and inclination charge, but it excludes multi-burn bielliptic options (tested and worse for small angles) and a plane change at higher altitude. Its time (16–36 d) is its real advantage, and a time-weighted objective could reverse the ranking.
3. *"15 m/s proximity operations and no phasing."* Correct, and decisive (Section 8). The count is conditional on that allowance.
4. *"Free insertion at the start object."* Yes; the starting-plane acquisition is not charged.
5. *"Counts are visits, not removals."* Yes; see Section 9.
6. *"Single pool, single epoch."* The pool is nearly exhausted at 5 km/s, and the SSO, Iridium-class and SL-16 bands appear in Table 3 only as medians.
7. *"Optimality gap not certified on the full pool."* Correct (Section 6).
8. *"Atmosphere, mass, kit and cost values are assumptions."* Yes. They are tagged in `research/params.json` (VERIFIED/DERIVED/UNVERIFIED); the headline numbers are insensitive to the atmosphere for this pool and sensitive to the proximity allowance.
9. *"The risk model is not validated."* It has no calibration to observed conjunction or fragmentation rates beyond the sanity check of 55 years between events if every derelict pair were involved, so it should be used for ranking only.
10. *"Was the literature search complete?"* No. Several papers named in the brief (Han 2024, Yang 2024 as stated, Ruggiero, Lüthi) could not be verified and were not used.

11 Conclusion

This work re-evaluated nodal-precession phasing for multi-object debris removal on one frozen catalogue snapshot, with an audit of the draft study, exact results on lowering versus raising, an exact-optimum check of the planner and high-fidelity verification of the leg model.

At the level of the phasing geometry, lowering gains over raising by $R(x) = 1 + \frac{9}{2}x + \dots$ in rate and by $1 + 3x$ per unit ΔV. The gain is small (4% at 100 km) and one-signed, so the bidirectional rule is better. At the level of the planner, a tuned beam search is within 2% of optimal on small instances.

At the mission level, nodal-precession phasing lets a single spacecraft visit a median of 29 SL-8 rocket bodies in 7 yr with 2 km/s, and 67 in 15 yr with 5 km/s, against 9 and 19 by impulsive plane change. This advantage is bought with time and depends on the unmodelled in-plane and proximity costs. A parked carrier with expendable kits is the one new-looking architecture that the numbers support. Everything else we tried was rejected or is prior art.

In general, the results demonstrate that drift phasing trades mission time for propellant rather than providing a time-free saving, and that the counts reported by the draft understated its own model while overstating what a removal mission could achieve. Future work should model the in-plane phasing and proximity-operations costs explicitly, size the kits of the carrier architecture as a complete design, and extend the analysis to other bands and to a refreshed catalogue.

Acknowledgments

The author acknowledges the public satellite catalogue and element-set services that made the frozen snapshot possible, and the authors of the open-source propagation and scientific Python libraries used throughout this analysis.

A Reproducibility Statement

Snapshot. v3-2026-08-18: Celestrak SATCAT csv (70,292 rows) and GP three-line sets (SL-8 159, SSO 135, Iridium-class 28, SL-16 20, mixed 393), public, no credentials, fetched 18 August 2026; the SHA-256 of every file is in `data/SNAPSHOT.json`.

Code. `run_all.py` runs the tests, audit, analysis, figures and this PDF; version hash `a3af289f888fd4c6` over 96 files (`CODE_MANIFEST.json`); Python 3.14, numpy 2.4, scipy 1.17, matplotlib 3.10, sgp4 2.25, sympy 1.14; no GPU, about 80 min on one core. This draft was prepared on 3 November 2026.

Seeds. Start subsets 7, 9, 42; Monte Carlo 1, 2026, 11, 100+replicate; NSGA-II 0–2; bootstrap 1–2; every stochastic step is seeded inside its script.

Provenance. The original repository was read-only; its code is copied to `legacy/` and used only to reproduce its numbers. No data were invented, and unverified inputs are listed in `research/params.json`.

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